

CTSpinoPelvic1K: a fused spine and pelvis CT segmentation dataset built for lumbosacral transitional anatomy

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Purpose: Lumbar levels cannot be numbered reliably when a lumbosacral transitional vertebra (LSTV) is present, because sacralisation and lumbarisation are one morphology under two counts and a spine-limited field of view cannot distinguish them. Public collections either omit the pelvis, or omit the classes a transitional vertebra requires: the largest whole-body collection has no L6 and no lumbar-rib class, and the reference spine benchmark excludes vertebrae fused to the sacrum by design. CTSpinoPelvic1K provides 802 CT records with spine, pelvis, per-level ribs and femora in one frame, in a scheme that can represent the variants, anchored at both ends of the lumbar column.

Acquisition and Validation Methods: Records were built by patient-level crosswalk of three public collections — TCIA CT COLONOGRAPHY imaging, CTSpine1K³ vertebral labels, and CTPelvic1K⁴ pelvic labels — with labels placed on the acquisition carrying the highest bone coverage and unified under a VerSe-native scheme. Validation is threefold: per-case geometric invariants (802/802 pass); rib-vertebra incidence across 5,749 evaluable ribs (2 residual offsets, 0.035%); and agreement of derived spinopelvic measures with published reference values.

Data Format and Usage Notes: Gzipped NIFTI image/label pairs sharing an identical affine, canonicalised to PIR, with frozen patient-grouped LSTV-stratified five-fold cross-validation splits and a reference loader. Archived at 10.5281/zenodo.XXXXXXX.

Potential Applications: Segmentation and level-numbering benchmarks, anatomical variant studies, surgical planning, and opportunistic screening research. *Limitations:* thoracic ground truth is field-of-view limited; postural angles are supine; soft-tissue and hardware identifiers are declared but unpopulated; the splits are cross-validation folds and include no held-out test set; the rib layer is a reviewed pseudolabel whose review was triaged by an automated rule rather than exhaustive; and the Castellvi grades are one reader's, with confirmation by a board-certified neuroradiologist in progress.

I. PURPOSE

Naming a lumbar vertebra requires counting, and counting requires two fixed ends. In the presence of a lumbosacral transitional vertebra neither end is where it is assumed to be: the same bone is a sacralised L5 or a lumbarised S1 depending only on where the count starts, and a field of view that shows the lumbar spine without the thoracolumbar junction cannot decide between them. LSTV is common — reported between 4% and 30% depending on definition — and wrong-level surgery is its most serious consequence.

Existing public collections do not make the problem tractable, and the reason is not that they are small (Table I). Spine collections annotate vertebrae without the pelvis, so the caudal anchor is missing; pelvic collections annotate the sacrum and ilia with the lumbar spine as a single undifferentiated class, so there is no count at all. The whole-body collections do annotate vertebrae, sacrum and ribs together, but their label schemes contain no L6 class, no class for a rib borne by a lumbar vertebra, and no transitional label — so a six-lumbar spine cannot

be written down in them at any segmentation quality. VerSe², which deliberately enriched for anatomical variants and did grade transitional vertebrae by Castellvi, states that it *did not segment* vertebrae showing partial fusion with the sacrum, Castellvi grades III and IV, and does not segment the sacrum. Those fused grades are 25 of the 33 graded records here.

CTSpinoPelvic1K exists to put all of it in one coordinate frame, in a scheme that can represent the variants rather than round them to the nearest normal.

The design commitment is that *the dataset never asserts a level name it cannot support*. Where the field of view cannot decide, the release reports count-free quantities: an interval count between the lowest rib and the sacrum, a rib length as a fraction of the rib above it, a transverse process span and its gap to the ala.

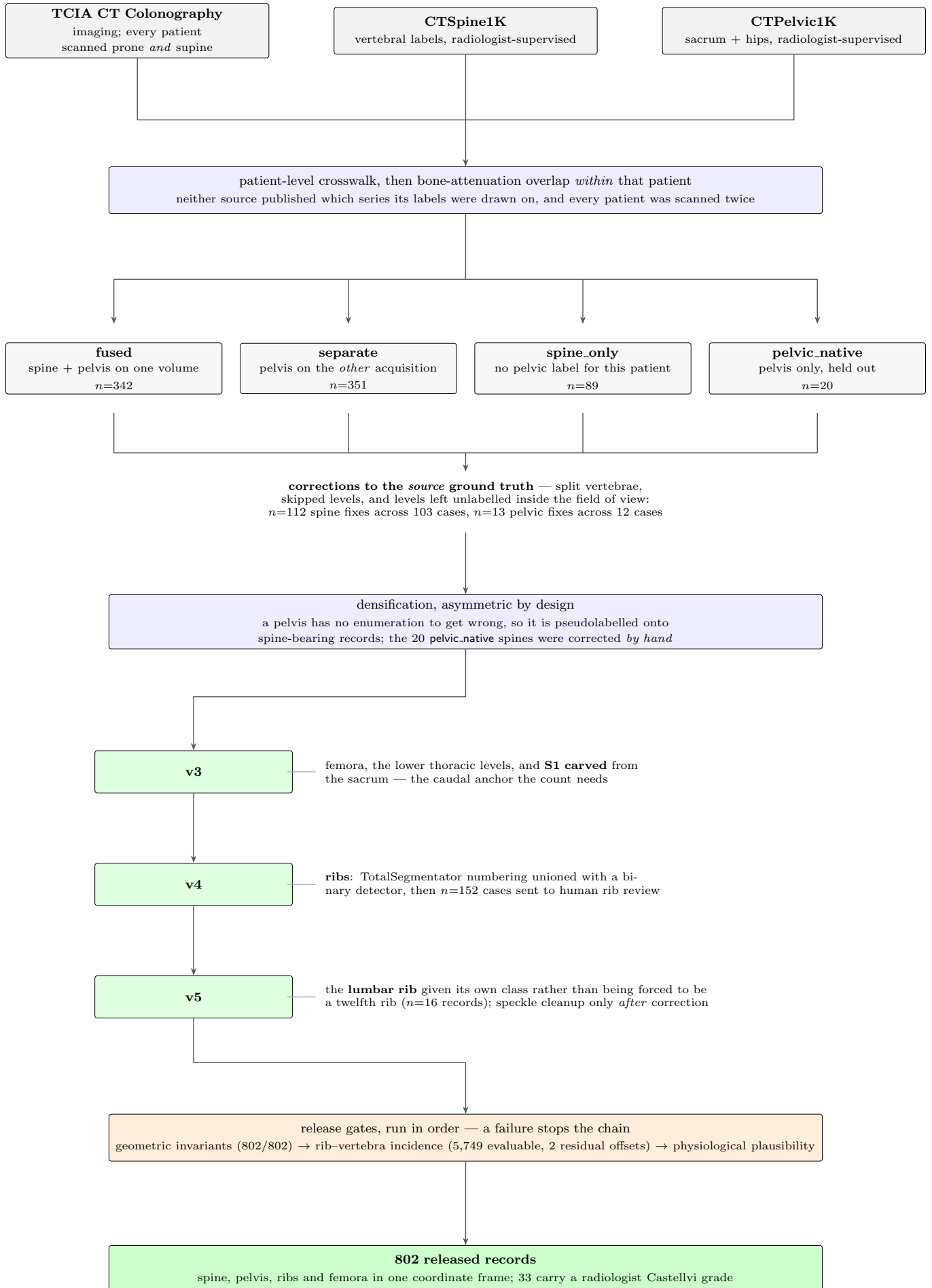


FIG. 1 Construction. Three public collections are joined at the patient level, and each annotation is then placed on the acquisition whose bone it actually agrees with — neither source published which series its labels were drawn on, and every patient in the colonography cohort

TABLE I Public CT collections covering the lumbosacral junction. Counts are scans as reported by each source. The columns after the first three are about what a label scheme can *represent*, not about segmentation quality: a collection with no L6 class cannot record a six-lumbar spine however well it segments. CTSpinoPelvic1K is not the largest of these and does not claim to be — TotalSegmentator and CTPelvic1K carry more scans, and RibSeg v2 labels more ribs. It is the only one in which the lumbosacral variants are expressible and graded. ^aVerSe grades transitional vertebrae by Castellvi but uses the grade as an *exclusion* criterion: vertebrae partially fused to the sacrum (grades III–IV) are not segmented, and the sacrum is not segmented at all. ^bThe lumbar spine is a single undifferentiated class, so no vertebral count is recoverable. ^cLabels produced by iterative pseudolabelling with manual refinement, rather than drawn under radiologist supervision.

Collection	scans	modality	vertebrae	L6	can the scheme represent ...			
					sacrum	pelvis	per-level ribs	lumbar rib
CTSpine1K ³	1005	CT	C1–L6	yes	no	no	no	no
CTPelvic1K ⁴	1184	CT	no ^b	no	yes	yes	no	no
VerSe ²	374	CT	C1–L6	yes	no ^a	no	no	no
RibSeg v2 ¹⁶	660	CT	no	no	no	no	yes	no
TotalSegmentator ¹⁴	1204	CT	C1–L5 ^c	no	yes	yes	yes	no
CTSpinoPelvic1K	802	CT	C1–L6	yes	yes (+S1)	yes	yes	yes

II. ACQUISITION AND VALIDATION METHODS

II.A. Sources, and why a crosswalk was necessary

Both CTSpine1K³ and CTPelvic1K⁴ draw part of their imaging from the TCIA CT colonography collection^{5,6}, and the vertebral and pelvic annotations in both were produced under board-certified radiologist supervision. Neither released the mapping from an annotation to the specific CT series it was drawn on.

That mapping is not a bookkeeping detail. Every patient in CT COLONOGRAPHY was scanned *twice* — prone and supine — and each acquisition was commonly reconstructed with more than one kernel, so a single patient identifier resolves to several volumes that differ in position, in field of view and in noise. An annotation carries a patient identifier and nothing finer. Placing it on the wrong volume of the same patient produces a mask that is plausibly shaped, correctly named, and anatomically wrong.

The crosswalk therefore proceeds in two stages. Each annotation is first resolved to a patient, and then, within that patient’s volumes only, to the series whose bone *agrees* with it: candidate volumes are thresholded at bone attenuation and scored by overlap with the annotation mask, and the annotation is assigned to the series that maximises it. Bone is used rather than image similarity because bone is what the annotation describes; two acquisitions of the same abdomen resemble each other closely everywhere except in where the skeleton sits.

II.B. Fused and split records

The two annotation sets were placed independently, so they do not always land on the same series, and the four resulting record types are distinguished because the distinction is the crosswalk’s main finding rather than bookkeeping. Of 802 records, 342 (42.6%) carry spine and pelvic annotations on the same volume and are exported as *fused*. A further 351 (43.8%) are *separate*: a pelvic annotation exists for the patient, but it was placed on the other acquisition of the prone/supine pair, so it

cannot be combined with the spine annotation without resampling across a change of posture. Only 89 (11.1%) are *spine-only*, meaning no pelvic annotation exists for that patient at all, and 20 (2.5%) are *pelvic-native*, carrying the pelvic annotation alone.

The 351 separate records are the direct measurement of the problem described above: in nearly half of all patients the two public collections annotated *different volumes of the same person*, and neither collection recorded which. They are retained rather than discarded, and held out of the postural analyses, because a prone and a supine acquisition of one patient differ in spinal alignment and are a resource for a mobility study rather than a defect.

II.C. Densification, and why it is asymmetric

A corpus in which half the records lack a pelvis is not usable as a spinopelvic dataset, so the missing structures were pseudolabelled. The two directions are not equivalent and were not treated as such.

Pelvis onto spine-only records. The sacrum and the innominate bones are single objects with no enumeration: there is no count to get wrong, no naming convention to choose, and no analogue of a transitional vertebra. A pelvic segmenter can therefore be trusted in a way a vertebral one cannot, and its quality is reported as held-out performance on the *pelvic-native* records, which were withheld from training for exactly this purpose.

Spine onto pelvic-native records. The reverse direction carries the whole problem this dataset exists to address. A pseudolabelled spine must commit to a vertebral count, and on a transitional vertebra it will commit to the commoner reading. Those 20 records were therefore not densified automatically: each pseudolabelled spine was reviewed and corrected by hand. Twenty cases is a tractable amount of manual work; four hundred would not have been, which is the practical reason the asymmetry runs the way it does.

II.D. Ribs

No public rib annotation covers this cohort, and the two available automatic sources fail in complementary ways.

Möller’s binary rib network¹⁵ is a detection model of high reported accuracy, but it segments ribs that lie *entirely within* the field of view; a rib clipped by the edge of an abdominal acquisition is liable to be missed altogether. Since this is a colonography cohort, the lower ribs are routinely clipped, and those are precisely the ribs the count depends on.

TotalSegmentator¹⁴ does segment clipped ribs and supplies the per-rib numbering that a binary network cannot. Its characteristic failure is at the other end of the bone: the most medial portion of the rib — roughly the last tenth to sixth approaching the costovertebral joint — is frequently absent. A rib truncated at its medial end never reaches the vertebra it articulates with, so the incidence test that assigns a rib to its level has nothing to test.

The two were therefore unioned: TotalSegmentator supplies numbering and the medial-clipped cases, Möller supplies detection where its output is more complete, and the union is carried forward with TotalSegmentator’s per-level identities.

The result is a pseudolabel, and the review it received was triaged rather than exhaustive — which is worth stating plainly, because a rib layer described only as “human-corrected” invites the reader to assume every bone was looked at. A label-based rule flagged every record in which one rib bone carried more than one identifier, or in which a rib failed to reach the vertebra it articulates with; a single bone with two numbers is a numbering error by construction, whatever the anatomy. That rule selected 152 records. 148 were reviewed and corrected by a trained reviewer; 4 were not, and are identified in the release. The remaining records passed the rule and were not individually inspected, so the guarantee the rib layer carries is that it is free of the failure modes the rule detects, and no stronger. The residual rib–vertebra offsets reported in Sec. II.F are the independent check on what the rule missed.

II.E. Label scheme and counting anchors

The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26), and every non-VerSe structure takes a fixed identifier above that range — S1 carved from the sacrum (29), hips (30–31), femora (32–33), ribs (34–45 left, 46–57 right), soft tissue (58–73), lumbar ribs (74–75) and hardware (76–79).

Two anchors make the count decidable. Rostrally, **T12** is the last rib-bearing vertebra and sits directly above ground-truth L1. Caudally, **S1** is carved as a distinct class, giving the bottom bracket and the sacral endplate landmark required for sacral slope and pelvic incidence.

With both brackets present, L5 and L6 are distinguishable where a spine-limited view alone would not be.

Neither anchor is novel on its own. TotalSegmentator¹⁴ also carries a separate S1 class alongside the sacrum and labels ribs per level, and VerSe² carries L6. What no published scheme carries is the pair of classes below.

A rib on a lumbar body receives its own class rather than being forced to be a twelfth rib. A lumbar rib and a hypoplastic twelfth rib are the same object under two counts; assigning it a number would bake one reading into the label. 16 records carry one. This is the class whose absence is decisive elsewhere: a scheme that numbers every rib 1–12 has nowhere to put a thirteenth, so the annotator must either call it rib 12 — which asserts the vertebra beneath it is thoracic, the very question at issue — or discard it.

L6 is a class, not an exception. A scheme without one cannot record a six-lumbar spine at all, and must renumber the column or drop a level to fit. The largest whole-body collection stops at L5.

Figure 2 shows why the count is the primitive and the name is derived. Of the 29 records with four rib-free vertebrae, 15 reach that count through a lumbar rib and 14 without one; the two are indistinguishable by the count alone and are distinguished here only because rib presence is itself labelled.

II.F. Validation

Validation is layered, and the ordering matters: measurements computed on a corpus that has not established internal consistency do not look broken, they look finished.

Geometric invariants. Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for left- and right-sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed. This check exists because a renumbering pass once wrote a reoriented array under a canonical affine and transposed one label away from its image — a failure invisible to every downstream step.

Rib–vertebra incidence. Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs present, 5,788 are not evaluable at all — the vertebra their number implies lies outside the field of view and nothing else is within the anchor distance — and a further 11 have no vertebra in reach. That leaves 5,749 with an anchor to check, of which two remain offset (0.035%). The denominator is stated because quoting two against 11,548 halves the rate by counting ribs the check never examined, and because more than half of the ribs in a screening abdominal CT being unevaluable is itself a fact about the corpus worth reporting. Both offsets are field-of-view truncations on individually characterised cases and are reported rather than suppressed: a threshold tuned until the count

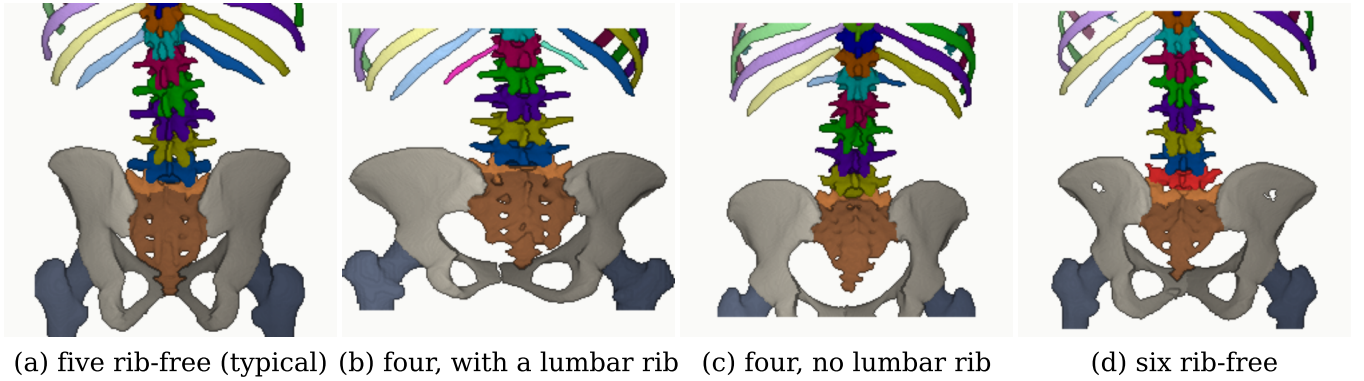


FIG. 2 The configurations the corpus exists to represent, rendered from the released labels at one shared posterior viewing angle. Columns are captioned by the number of rib-free vertebrae between the lowest rib-bearing vertebra and the sacrum, not by a level name: a spine-limited field of view cannot settle whether (b) is a sacralised L5 or (d) a lumbarised S1, and the count can be checked against the picture where a name cannot. Panels (b) and (c) carry the same count, reached from opposite ends — in (b) a rib on the first lumbar-type vertebra shortens the interval from above, in (c) the lowest lumbar segment is incorporated into the sacrum and shortens it from below. Surfaces and colours are those of the released label descriptor, so the claims are checkable against the image: the lumbar rib in (b) carries its own class and renders magenta, and the sixth lumbar vertebra in (d) renders red directly above the sacrum. Each column is cropped to a fixed height above the top of the sacrum rather than to a fraction of the labelled spine, so the columns are framed on the same anatomy and a taller patient shows fewer levels rather than a rescaled skeleton.

reaches zero yields a cleaner number and a less informative one.

Agreement with published values. Derived spinopelvic measures are compared against Vialle *et al.*¹ (Table IV). Pelvic incidence is the strongest of these checks because it is a morphological property of the pelvis rather than a posture, and so is directly comparable to a standing reference cohort.

II.G. Castellvi typing, and the reading it still requires

Every record whose vertebral count departs from five carries a Castellvi grade⁹ in the released manifest: 33 cases, typed I–IV with the *a/b* unilateral–bilateral qualifier (Table III). This is released as an annotation layer because the grade and the count are *different axes* and the dataset is built to show that they are: grade IIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a perfectly normal count of five. A corpus that recorded only the count would describe those seven as unremarkable.

The grading is a single reader’s, with five cases independently read a second time. Three of those five agreed exactly. Of the two that did not, one is Ib against IIb — adjacent subtypes, the ordinary kind of disagreement — and the other is IIb against IIb, which crosses the boundary between bony fusion and articulation without it. That is the distinction the classification is built on and the one carrying clinical weight¹³, so it is reported rather than folded into a single agreement rate. Five second reads establish that the grading was checked; they are far too few to support an inter-rater statistic, and none is quoted.

The typing is therefore submitted for confirmation by a board-certified neuroradiologist, blinded to the existing grades and to the vertebral counts, reading all 33

cases. Where the confirming read and the original agree the grade stands; where they disagree the case is adjudicated in a joint reading and the adjudicated grade released, with both original reads retained in the manifest so that no disagreement is silently resolved. Full agreement statistics will be reported for the complete set of 33 rather than for the current subset of five. *This confirmation is in progress and the release is not final until it completes*; the grades described here are the single-reader grades, and the manifest records which reads each grade rests on.

III. DATA FORMAT AND USAGE NOTES

Each record is a gzipped NIFTI image/label pair sharing an identical 4×4 affine, canonicalised to PIR, requiring no resampling. Masks are NIFTI rather than DICOM: the policy prefers DICOM where applicable, and for volumetric segmentation masks NIFTI is what the downstream tooling expects.

Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the data as `splits.5fold.json`. The five validation folds partition all 802 records — they are disjoint and their union is the cohort — so the release provides cross-validation and *not* a held-out test set. Anyone reporting a single held-out number should carve one out and say which records it holds, because the folds shipped here will not do that job.

Stratification is on the transitional subtype rather than on a binary LSTV flag, and it enforces a minimum of each rare subtype in every validation fold: each fold’s validation set carries three sacralisation and three lumbarisation records, so no fold is evaluated on a cohort missing the class the dataset exists for. With 15 lumbarisation and 15 sacralisation records across 802, that is the prac-

tical limit of what stratification can guarantee, and it is the reason a fold-level metric on the rare classes has an error bar far wider than its own decimal places.

The archive of record is Zenodo (10.5281/zenodo.XXXXXXX). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM’s *other* value for patient sex, which is a recorded value and not a missing one, and 53 carry no sex field at all. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals, and records outside the female/male dichotomy are excluded from measures stratified by sex — a limitation of those measures rather than a property of the cohort, and stated here because folding eleven people into “unrecorded” would misdescribe what the source says. The lower age bound is the screening guideline rather than a selection applied here, and it is why the measures below sit closer to a reference range than a surgical series would.

Count-free measures (Fig. 3). Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal, with modes near 0.33 and 0.69.

Level gradients and spinopelvic measures (Fig. 4). Vertebral dimensions increase caudally in step with the load each level carries. Pelvic incidence does not vary with age, while pelvic tilt increases — the descriptive signature of a pelvis retroverting as a spine ages.

Opportunistic measures (Fig. 5). Every scan carries a vertebral bone density measurement at no additional dose. L1 trabecular attenuation is reported at the published standard site.

V. POTENTIAL APPLICATIONS AND LIMITATIONS

The dataset supports segmentation and level-numbering benchmarks, anatomical variant studies, surgical planning research, and opportunistic screening work of the kind Fig. 5 illustrates.

Its limitations are part of it, and they are stated at the level of detail a reader would need to catch them independently.

Coverage. Thoracic ground truth is field-of-view limited and does not extend to T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat. The cohort is a colorectal screening population aged 50 and over, so its distributions should not be read as representative of a surgical one.

Declared but empty. Soft-tissue (58–73) and hardware (76–79) identifiers are declared in the scheme and popu-

lated by no case in this release. They are retained so that a downstream annotation can use them without renumbering, and a user should treat their absence as absence of annotation rather than absence of the structure.

Strength of the labels varies by structure, and the release does not average over that. Vertebral labels derive from radiologist-supervised source annotations, corrected where those sources were wrong. Pelvic labels on records that lacked one are pseudolabelled. The rib layer is a pseudolabel whose human review was triaged by an automated rule rather than exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional labels derive from two independent sources and are not uniformly adjudicated — which is precisely why the measures reported here are count-free — and the Castellvi grades are a single reader’s pending the confirmation described in Sec. II.F.

Splits. The shipped folds are cross-validation folds covering the whole cohort. There is no held-out test set, and with 15 lumbarisation and 15 sacralisation records a per-fold metric on the rare classes is estimated from three cases.

DATA AVAILABILITY

The dataset is permanently archived at <https://doi.org/10.5281/zenodo.XXXXXXX>. Code is at [repository URL], tagged at the release commit.

CONFLICT OF INTEREST

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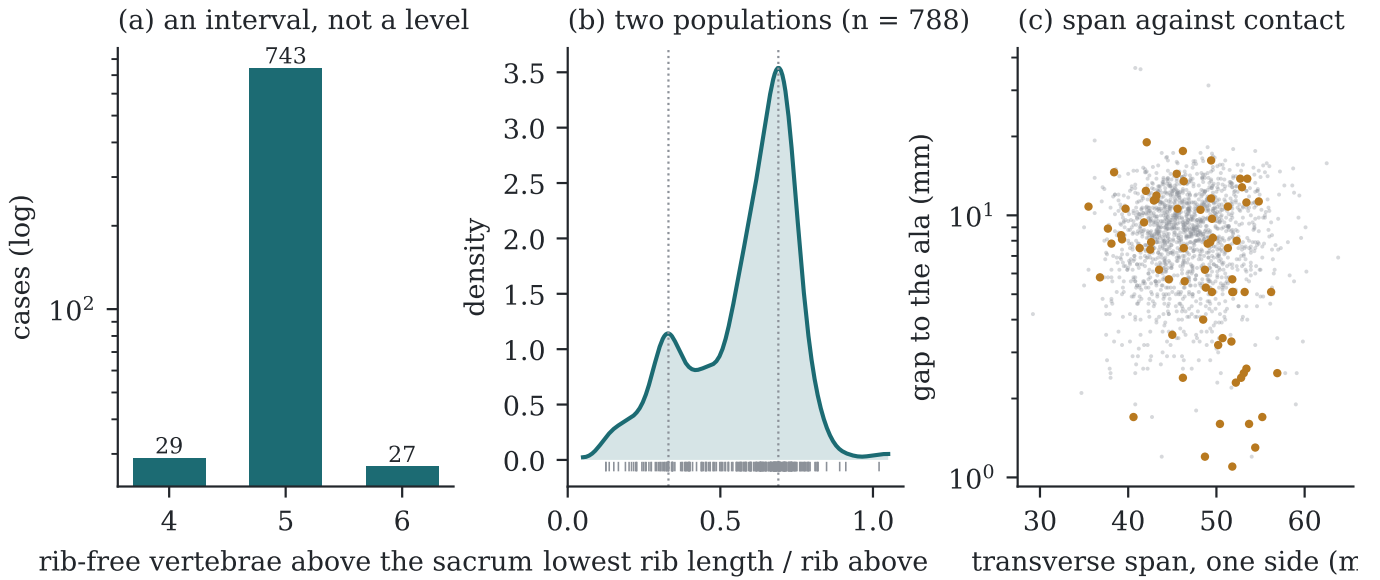


FIG. 3 Count-free measures, valid without knowing which vertebra is which. (a) Vertebrae between the lowest rib and the sacrum: an interval, not a level assignment. (b) Lowest rib length as a fraction of the rib above, showing two populations near 0.33 and 0.69 rather than one distribution with a tail. (c) Lowest lumbar transverse span against its gap to the sacral ala; cases carrying a source transitional label are marked.

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TABLE II Release validation. Gates run in order and a failure stops the chain.

Check	Result
Geometric invariants (all cases)	802/802 pass
label shape, affine, spacing match CT	—
no identifier outside the scheme	—
left/right structures correctly sided	—
Ribs present	11,548
not evaluable (expected vertebra outside the FOV)	5,788
no vertebra within the anchor distance	11
evaluable	5,749
matching the expected vertebra	5,747
residual offset	2 (0.035% of evaluable)
misnumbered cases	0
Spinopelvic medians within published range	6/6

TABLE III Radiologist Castellvi grade against the rib-free vertebral count, for the 33 graded records. The two are different axes and the table shows how far apart they run: grade IIIb occurs at counts of four, five *and* six, and seven graded records carry the normal count of five, where nothing about the count would draw attention.

Castellvi grade	records	rib-free vertebrae		
		4	5	6
Ib	2	0	0	2
IIa	2	0	1	1
IIb	4	0	0	4
IIIa	3	0	1	2
IIIb	18	7	4	7
IV	4	1	1	2
all	33	8	7	18

TABLE IV Derived measures against a published asymptomatic reference ($n = 260$, standing)¹. Sacral slope and pelvic tilt are postural where pelvic incidence is not, and this cohort is supine. The two postural measures depart from the standing reference in opposite directions — sacral slope 4.1° lower, pelvic tilt 4.3° higher — which is not two discrepancies but one, because $PI = PT + SS$ holds by construction: with pelvic incidence fixed at its published value, any fall in sacral slope must appear as an equal rise in pelvic tilt. That the two offsets match in magnitude is evidence the geometry is internally consistent rather than evidence of error.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	54.7 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis (supine)	52.7°	43 ± 11.2 (standing)
PI – LL mismatch	2.6°	centred near 0

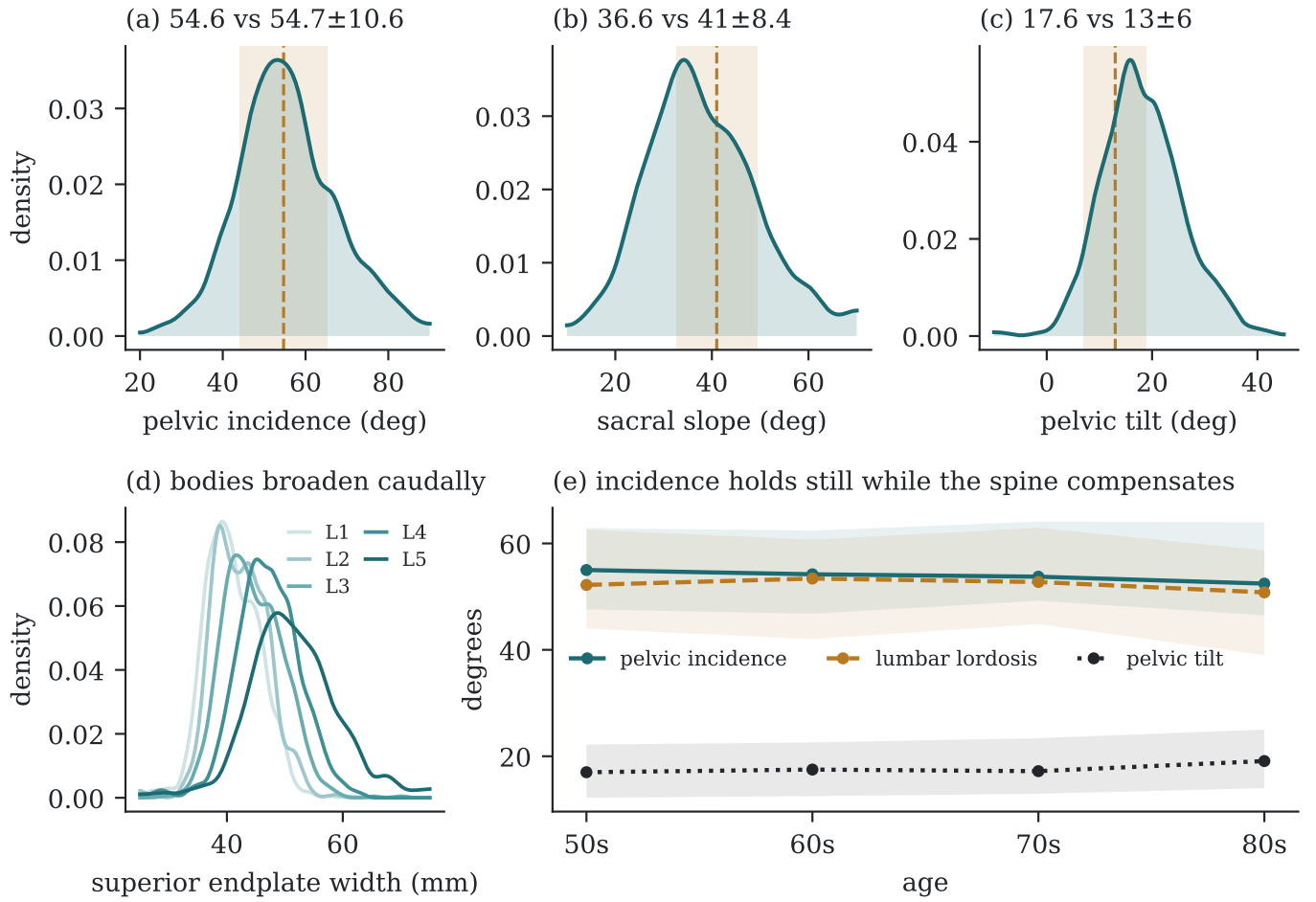


FIG. 4 (a-c) Spinopelvic measures against published reference bands (mean $\pm$ SD). (d) Superior endplate width by level. (e) Median and interquartile range by decade; pelvic incidence is flat by construction of the anatomy, which makes it a negative control for the measures beside it.

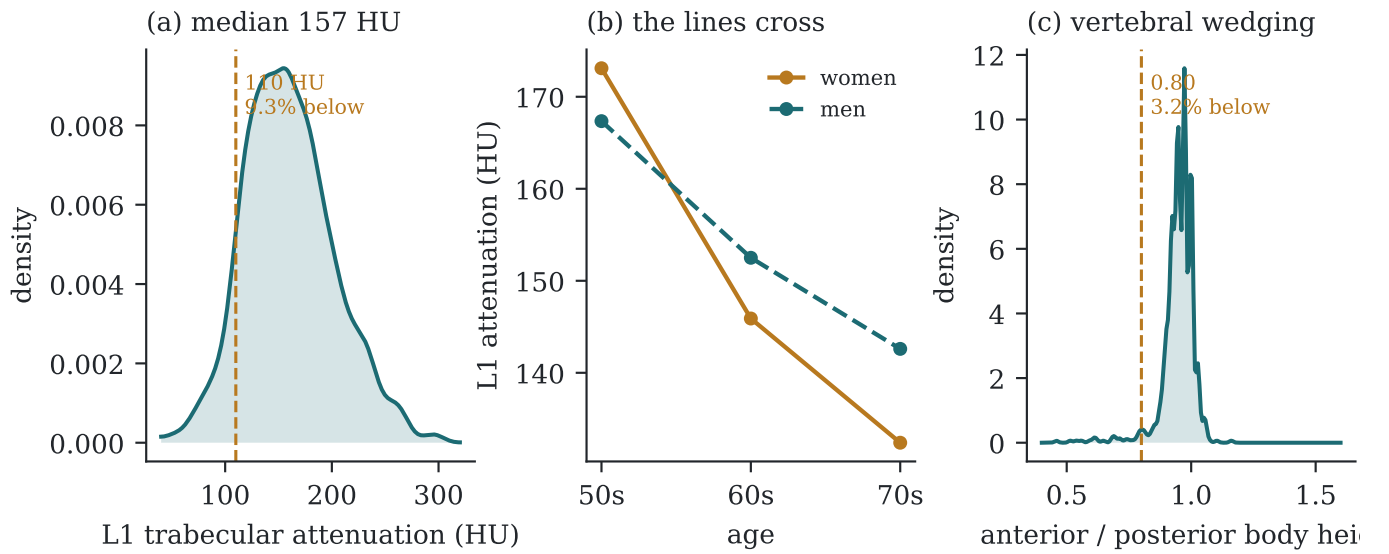


FIG. 5 Opportunistic measures from the same images. (a) L1 trabecular attenuation with the 110 HU osteoporosis-specific threshold. (b) Median attenuation by decade and sex. (c) Lowest lumbar wedge ratio with Genant⁸'s 0.80 threshold for mild wedge deformity.